

CMP-24 - Observability and Audit Platform

Collect structured logs, metrics, traces, Audit Records, health evidence candidates, and tamper-evident audit chronology without owning Evidence Objects.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 25. Authoritative state: Audit Ledger, Audit Records, telemetry pipelines, metric and trace stores, alert state. Explicitly non-owned: Evidence Objects, domain authoritative state, Trust Assessments, incident command.

Critical invariant

Observability may produce Evidence Candidates, but only Evidence Service admits Evidence Objects or changes evidentiary meaning.

Logical interfaces

CMP-24-IF-01 / Command / Append Audit Record

Command `Append Audit Record` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-02 / Command / Emit Telemetry

Command `Emit Telemetry` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-03 / Command / Declare Alert

Command `Declare Alert` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-04 / Command / Seal Audit Segment

Command `Seal Audit Segment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-05 / Query / Query Audit Ledger

Query `Query Audit Ledger` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-06 / Query / Get Trace

Query `Get Trace` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-07 / Query / Get Metric

Query `Get Metric` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-08 / Query / Verify Audit Chain

Query `Verify Audit Chain` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-09 / Event / Audit Record Appended

Event `Audit Record Appended` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-10 / Event / Alert Raised

Event `Alert Raised` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-24-IF-11 / Event / Audit Chain Broken

Event `Audit Chain Broken` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Healthy > Degraded > Buffering > Recovering > Healthy. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: sensitive logging, audit gap, trace spoofing, telemetry loss, ledger tampering. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-24-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-24-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-24-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-24-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-24-TST-005 - Critical invariant. Observability may produce Evidence Candidates, but only Evidence Service admits Evidence Objects or changes evidentiary meaning. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-24-TST-006 - Failure containment. Inject sensitive logging, audit gap, trace spoofing, telemetry loss, ledger tampering. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-24-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-24-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-24-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-24-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.